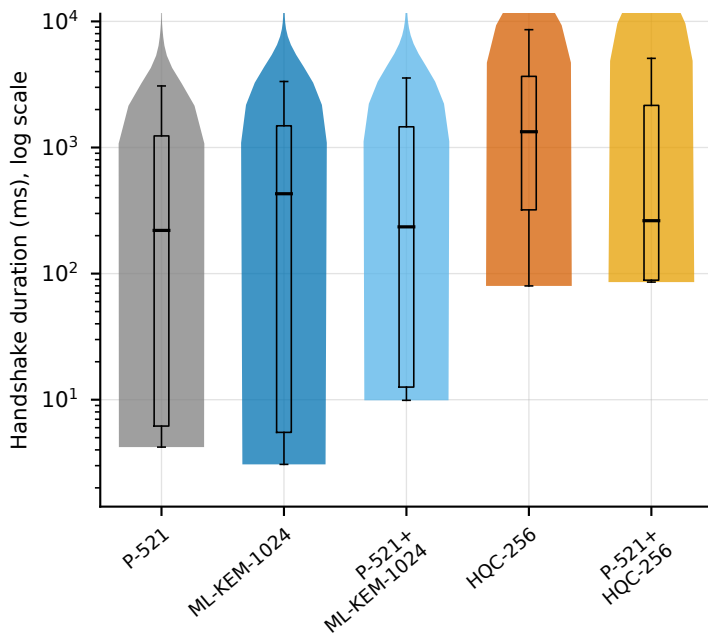
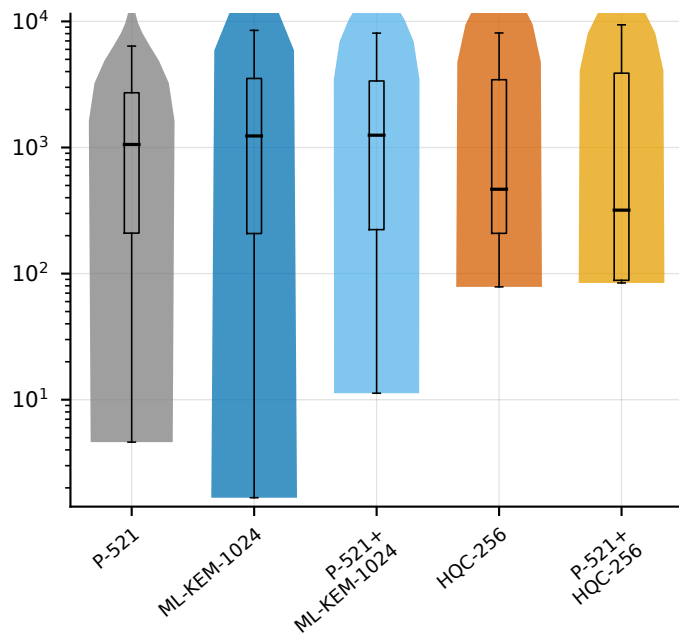


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	1.64k	391.4	105.88k	3.74k	310.7	160.41k
ML-KEM-1024	2.33k	377.1	107.61k	5.91k	491.1	583.46k
P-521+ML-KEM-1024	2.20k	413.8	109.52k	7.07k	410.5	341.34k
HQC-256	7.24k	434.5	457.82k	7.24k	480.6	463.26k
P-521+HQC-256	7.28k	559.6	472.97k	7.26k	403.9	393.47k